Lesson 1.12: Transformations of Functions

Topic 1.12: Transformations of Functions

A set of functions that form a family, namely, $F(x) = a \cdot f(b(x + h) + k$, is commonly called the **parent function. A change made to at least one of the values** a, b, h and k, is called a **transformation.** There are four types of transformations: ***dilation, reflection, rotation,*** and ***translation.***

Rigid Transformations:

Keep the original shape and size of the graph unchanged

Notation	Changes to Parent Function	Ordered Pair
$g(x) = f(x) + k$	Vertical translation $f(x)$ k units	$(x, y+k)$
$g(x) = f(x + h)$	Horizontal translation $f(x)$ $-h$ units	$(x-h, y)$
$y = -f(x)$	Reflect function across x-axis	$(x, -y)$
$y = f(-x)$	Reflect function across y-axis	$(-x, y)$

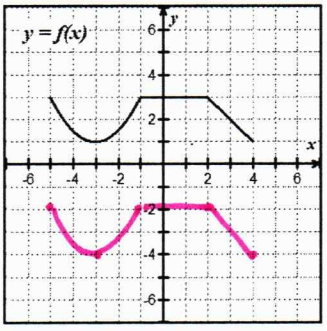
Important Observations

- ❑ Transformations that occur **inside the parentheses** affect the ***x*-values** of the graph; and **do the opposite of the expression.**
  For example: $f(x + 4)$ means a transformation in the opposite direction $x = -4$ or left.
- ❑ Transformations that occur **outside the parentheses** affect the ***y*-values** of the graph.

Vertical Translation

$g(x) = f(x) - 5$

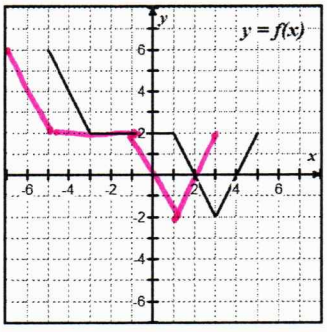
down 5



Horizontal Translation

$g(x) = f(x + 2)$

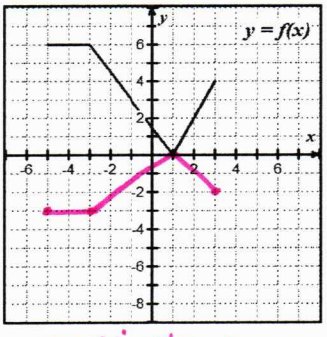
left 2



Vertical Dilation

$g(x) = -2f(x)$

reflect ↓ compress

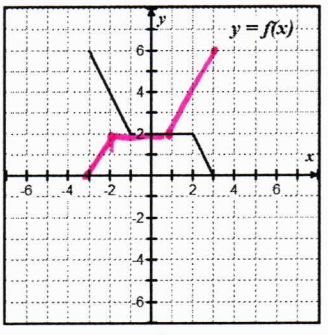


multiply y

Horizontal Dilation

$g(x) = f(-x)$

reflect about y



multiply x

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Multiplicative Transformations:

Non-Rigid Transformations distort the shape of a graph

Notation	Changes to Parent Function	Ordered Pair
$y = a \cdot f(x);\ \lvert a\rvert > 1$	Vertical dilation (stretch) factor of $\lvert a\rvert$	$(x,\ ay)$
$y = \frac{1}{a} \cdot f(x); 0 < \lvert a\rvert < 1$	Vertical dilation (shrink) factor $\lvert a\rvert$	$(x,\ ay)$
$y = f(b \cdot x);\ \lvert b\rvert > 1$	Horizontal dilation (shrink) factor $\left\lvert\frac{1}{b}\right\rvert$	$\left(\frac{x}{b},\ y\right)$
$y = f\left(\frac{1}{b} \cdot x\right); 0 < \lvert b\rvert < 1$	Horizontal dilation (stretch) factor $\lvert b\rvert$	$\left(\frac{x}{b},\ y\right)$

EX #2: The table below gives values of the function k at selected values of x.

Let $h(x) = 3k\left(\frac{1}{2}x\right) + 2$. Find the following values:

x	-5	-3	-1	0	1	3	5
$k(x)$	-3	0	1	4	3	-1	-4

A. $h(2) = 3k(1) + 2$
$= 3(3) + 2$
$= 11$

B. $h(0) = 3k(0) + 2$
$= 3(4) + 2$
$= 14$

C. $h(10) = 3k(5) + 2$
$= 3(-4) + 2$
$= -10$

D. $h(-6) = 3k(-3) + 2$
$= 3(0) + 2$
$= 2$

EX #3: The domain of a function $y = f(x)$ is $-6 \le x \le 8$ and the range of f is $-3 \le y \le 7$.

Find the domain and range of h, where $h(x) = 2 \cdot f(x+1) - 3$.

shift left 1 $(x-1)$

domain of f: $[-6, 8]$

domain of h: $[-6-1,\ 8-1]$

$[-7, 7]$

range of f: $[-3, 7]$

range of h: $[2(-3)-3,\ 2(7)-3]$

$[-9, 11]$

vertical stretch by 2 ↓ shift down 3 $(2y-3)$

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EX #4: The function $f(x)$ has a domain of $[-5,5]$ and a range of $[-15,15]$. The function g is represented by the equation $g(x) = -2f\left(\frac{1}{3}(x+3)\right) + 7$. Describe the transformations. Then find the domain and range of $g(x)$.

Horizontal dilation by 3 and shift left 3

domain of $f(x)$: $[-5,5]$, domain of g: $[3(-5)-3, 3(5)-3] = [-18,12]$

Vertical dilation by 2, reflect across x-axis, shift 7 up

range of $f(x)$: $[-15,15]$, range of g: $[-2(-15)+7, -2(15)+7] = [23,37]$

We can also use the transformation rules for polynomial functions as a rapid method to write expressions for the transformation or find values of a transformed function.

x	-6	-4	-2	-1	0	2	3
$f(x)$	65	51	12	-3	-6	4	15

EX #5: The table above gives values for a polynomial function f at selected values of x. Let $g(x) = af(bx) + c$, where a, b, and c are positive constants. In the xy-plane, the graph of g is constructed by applying three transformations to the graph of f in this order:

A horizontal dilation by a factor of 3, a vertical dilation by a factor of 2, and a vertical translation by 5 units. Find the value of $g(-6)$. $b = \frac{1}{3}$, $a = 2$, $c = 5$

$g(x) = 2f\left(\frac{1}{3}x\right) + 5$

$g(-6) = 2f\left(\frac{1}{3}(-6)\right) + 5$

$= 2f(-2) + 5$

$= 2(12) + 5 = 29$

EX #6. The function f is not explicitly given. In the xy-plane, the graph of h is created by a sequence of transformations on f. The graph of h is the result of a vertical dilation of the graph of f by a factor of $\frac{1}{2}$, then dilating horizontally by a factor $\frac{1}{3}$, then translating horizontally to the left 7 units and finally translating vertically 3 units down. What is the equation for $h(x)$ in terms of $f(x)$? $a = \frac{1}{2}$, $b = 3$, $h = +7$ $k = -3$

$h(x) = \frac{1}{2} f[3(x+7)] - 3$

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